

Estimation of Streambed Groundwater Fluxes Associated with Coaster Brook Trout Spawning Habitat

by Matthew Van Grinsven¹, Alex Mayer², and Casey Huckins³

Abstract

We hypothesized that the spatial distribution of groundwater inflows through river bottom sediments is a critical factor associated with the distribution of coaster brook trout (a life history variant of *Salvelinus fontinalis*) spawning redds. An 80-m reach of the Salmon Trout River, in the Huron Mountains of the upper peninsula of Michigan, was selected to test the hypothesis based on long-term documentation of coaster brook trout spawning at this site. A monitoring well system consisting of 22 wells was installed in the riverbed to measure surface and subsurface temperatures over a 13-month period. The array of monitoring wells was positioned to span areas where spawning has and has not been observed. Over 200,000 total temperature measurements were collected from five depths within each monitoring well. Temperatures in the substrate beneath the spawning area were generally less variable than river temperatures, whereas temperatures under the nonspawning area were generally more variable and closely tracked temporal variations in river temperatures. Temperature data were inverted to obtain subsurface groundwater velocities using a numerical approximation of the heat transfer equation. Approximately 45,000 estimates of groundwater velocities were obtained. Estimated groundwater velocities in the spawning area were primarily in the upward direction and were generally greater in magnitude than velocities in the nonspawning area. Both the temperature and velocity results confirm the hypothesis that spawning sites correspond to areas of significant groundwater flux into the river bed.

Introduction

Coaster brook trout are a unique life history variant of the brook trout species, *Salvelinus fontinalis*, which have been documented in Lake Superior for hundreds of years (Newman and Dubois 1996; Huckins et al. 2008). Although populations of coaster brook trout were once

common in coastal tributaries throughout the Lake Superior basin, the Salmon Trout River is the only river on the southern shore of Lake Superior (Figure 1) that supports a verified, naturally reproducing remnant population of adfluvial coaster brook trout (Huckins et al. 2008). A long-term ecological study of coaster brook trout in the Salmon Trout River has investigated their adfluvial patterns, spawning density and distribution, behavioral characteristics, and genetic linkages (Huckins and Baker 2008; Huckins et al. 2008). To support conservation efforts within the Salmon Trout River and rehabilitation efforts along the southern shore of Lake Superior, habitat conditions (e.g., characteristics of spawning habitat) associated with a naturally reproducing coaster brook trout populations need to be characterized.

The spatial distribution of groundwater discharges can have a significant impact on the distribution of benthic

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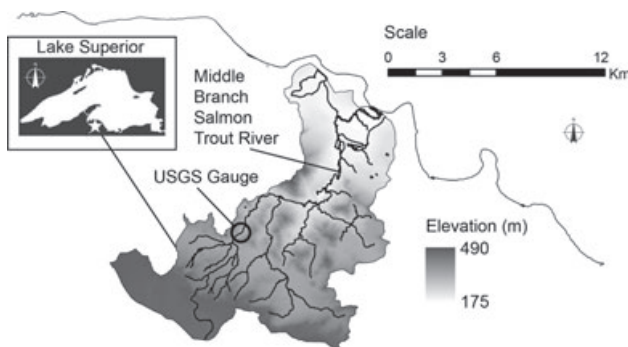


Figure 1. Salmon Trout River watershed map, Marquette County, Michigan.

and hyporheic fauna (Brunke and Gonser 1997; Storey et al. 2003) and the selection of spawning locations by fish (Curry and Noakes 1995). Based on relatively short-term and coarse estimates of groundwater seepage (e.g., from seepage meters, mini-piezometers, or direct temperature measurements), several studies have shown a relationship between brook trout spawning habitat and groundwater seepage (Fraser 1982; Curry and Noakes 1995; Ridgeway and Blanchfield 1998).

Groundwater seepage in permeable substrates functionally stabilizes critical biological water quality parameters such as temperature and oxygen availability in near-subsurface riverine environments (Fraser 1985; Curry et al. 1995). Consistency of thermal and chemical properties is critical for the survival of developing embryos in climates which undergo drastic seasonal changes (Curry et al. 1995), as exemplified in cold northern regions where formation of benthic ice layers are common during the winter months.

Various techniques have been implemented in order to determine subsurface fluxes in stream beds, including estimates of differences between upstream and downstream discharge over lengths of stream bed using current meters (Becker et al. 2004), pressure-head measurements acquired from in-stream monitoring wells (Curry et al. 1994), chemical tracer injection experiments (Constantz et al. 2003), seepage meters (Blanchfield and Ridgeway 1996; Alexander and Caissie 2003), and acquisition of vertical temperature profiles beneath streams (Lapham 1989; Silliman et al. 1995; Stonestrom and Constantz 2003; Constantz 2008).

Spatial patterns and magnitude of groundwater discharge can be critical when aquifer-stream transport and fate of contaminants is an issue (Conant 2004). Temporal changes in water table and stream levels can alter the rate and direction of subsurface fluxes under stream beds (Wroblicky et al. 1998).

Vertical temperature profiles can be used to produce quantitative estimates of groundwater fluxes into or out of streambeds at specific aerial locations and as a function of time. The procedure involves measuring temperature at multiple points below the streambed, where the temperature gradient is assumed to be a result of the heat transport processes (advection and

conduction). Heat transport models are calibrated to temperature observations and the groundwater flux is inverted from the calibrated model (Anderson 2005; Hatch et al. 2006). Minimum fluxes between approximately 10^{-2} cm/s (Silliman et al. 1995) and approximately 10^{-5} cm/s (Schmidt et al. 2006) have been estimated using this procedure. Although groundwater flux estimates are quantitative, they are subject to a number of assumptions concerning the dimensionality of the system, boundary conditions, and spatial distributions of the thermal characteristics of the streambed and aquifer material; (Becker et al. 2004; Ferguson and Bense 2011; Schornberg et al. 2011; Shanafield et al. 2011).

In this study, high-resolution temperature data collection methods were implemented to quantify the interaction between groundwater and surface water in order to verify the presence or absence of groundwater discharge in the river at sites that support a reproducing population of coaster brook trout. Monitoring wells equipped with vertically stratified temperature sensors were installed throughout a contiguous reach of river that contained an area known to be actively used by coaster brook trout for spawning and areas not known to support spawning. Temperature measurements were collected over a 13-month period (2007 to 2008). A numerical solution to the one-dimensional convective-diffusive heat transport equation was inverted to obtain best fit estimates of vertical groundwater fluxes throughout the study period. We hypothesize that the spatial distribution of groundwater inflows through river bottom sediments is a critical factor associated with the distribution of coaster brook trout spawning redds and activities.

Methods

Study Site

The Salmon Trout River watershed drains approximately 12,690 ha of subboreal forestland and descends approximately 245 m in elevation between the headwaters and outlet (Bullen 1988). Average annual precipitation in the area is 762 mm (NCDC 2009, snowfall is measured as liquid equivalent). Average annual snowfall is 2,900 mm, contributing to large spring snowmelt discharge events, while relatively dry summers produce low-discharge, baseflow-dominated periods (NCDC 2009). An average annual discharge of $0.16 \text{ m}^3/\text{s}$ was recorded over the period 2004 to 2008 (USGS 2009). The gauging station is located approximately 15 km upstream from the study site, and covers approximately 32% of the area drained to the point of the study site (Figure 1). Temperatures measured 10 m downstream of the study site in the Salmon Trout River vary between 0°C in the winter and 18 to 24°C in the summer (Figure 2). Several documented coaster brook trout spawning sites in the Salmon Trout River (Marquette County, Michigan) occur within approximately 12 river kilometers upstream from the river's Lake Superior outlet (Huckins and Baker 2008). This section of the river is characterized by a

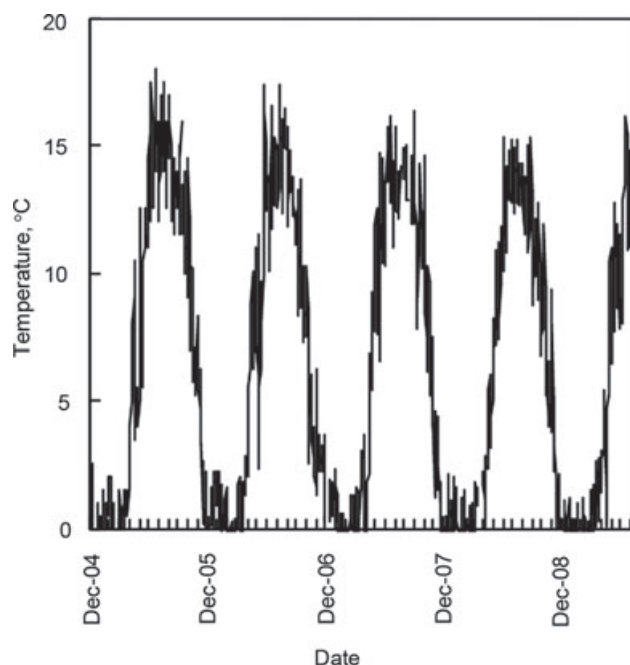


Figure 2. Temperature as a function of time in the Salmon Trout River. Source: Casey Huckins, Michigan Technological University, unpublished data.

relatively shallow gradient, unconsolidated bed material, wide active channels, and deep cut bank overhangs.

The study site is located along a gradual meander in the river several river kilometers upstream from the mouth. The bed contains poorly sorted materials consisting of cobbles, gravel, and sand. The site is characterized by environmental habitat features that have been previously correlated to suitable salmonid spawning habitat, including the occurrence of in-stream coarse woody debris, dense riparian vegetation, under cut banks, and coarse bed materials (e.g., Curry and Noakes 1995; Bernier-Bourgault and Magnan 2002; Kondolf et al. 2008). It was determined that environmental habitat conditions within the study site were sufficiently similar to permit isolation of groundwater flux as an independent variable associated with the location of naturally re-occurring spawning habitat by the coaster brook trout.

A recent ecohydrologic study (Kolka et al. 2010) conducted at a nearby site can be used to infer the general hydrogeologic characteristics of the subject site. Floodplains in local riparian areas are characterized as sandy stream terraces with deep, moderately well-drained loamy sands with hydraulic conductivity values that range from 4×10^{-4} to 1×10^{-1} cm/s. Depths to the water table range from less than 1 to 2 m below the surface at the top of the hillslopes, with horizontal hydraulic gradients on the order of 0.001 extending from the top of the hillslopes to adjacent streams. Water table elevations fluctuate on the order of less than 0.7 m over a typical year.

Network Design and Instrumentation

A contiguous network of nine evenly spaced transects each containing three monitoring wells was installed into

an 80-m section of the river. Each transect included one monitoring well located on the in-stream side of each bank, and one at the midpoint between the two river banks. Prior to installation of the monitoring wells, the study site was separated into an active spawning section and a nonspawning section according to the observed locations of active redd building recorded during the 2007 and 2008 spawning seasons (Figure 3). Location of bank-side monitoring wells permitted the collection of temperature data during active spawning periods. Each monitoring well consisted of a 1.8 m-long section of schedule 40 PVC pipes with an inner diameter of 3.8 cm and a 1.5 m-long screen with 0.25 mm slot openings (Figure 4). The

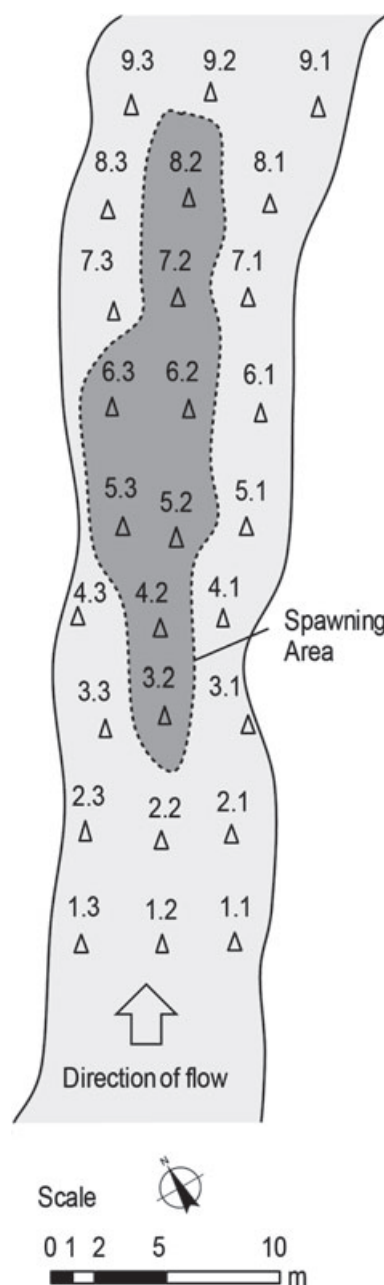


Figure 3. Locations of monitoring wells and areas where coaster brook trout have been observed to spawn consistently.

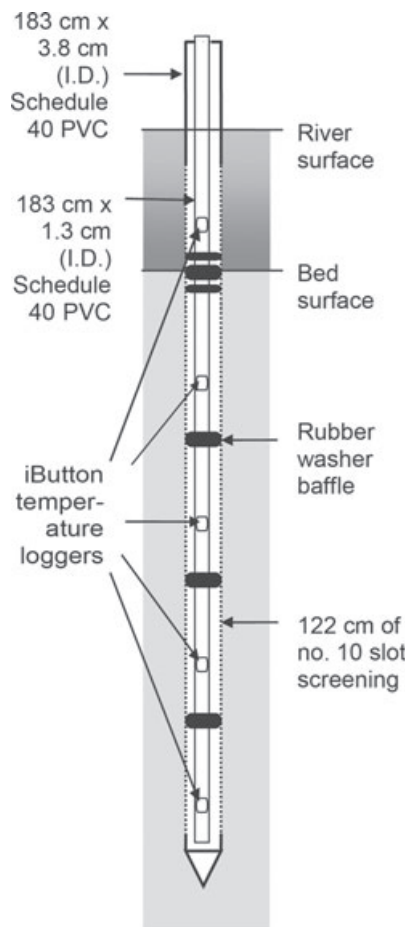


Figure 4. Monitoring well instrument configuration.

monitoring wells were installed using a rod and casing apparatus to depths of approximately 1 m beneath the bed surface. Each monitoring well was developed using a surge block technique.

Temperatures were measured with Thermochron iButton (Embedded Data Systems, Lawrenceburg, KY) temperature loggers capable of recording temperatures between -5°C and 26°C with a resolution of 0.125°C and an accuracy of 0.1°C . Five loggers, spaced approximately 24 cm apart in the vertical direction, were attached to a 1.25 cm diameter schedule 40 PVC pipes, and installed in each monitoring well (Figure 4). Each temperature assemblage consisted of one logger installed at a height corresponding to approximately 12 cm above the river bed, with the four remaining loggers located beneath the river bed. Depths of the temperature loggers for each monitoring well are reported in Table 1. In order to prevent vertical mixing within monitoring wells, SantopreneTM rubber washer baffles were attached to the PVC pipe at the midpoint between individual loggers. In addition, the uppermost baffle consisted of three staggered rubber washers, to ensure exchange between the surface and subsurface water did not occur. Data collection began prior to the 2007 spawning season (October) and continued through the duration of the 2008 spawning season (November).

Table 1
Installed Depths of Temperature Loggers

Monitoring Well	Temperature Logger Depth (cm Below River Bed)				
	1	2	3	4	5
1.1	-15	14	43	72	101
1.2	-9	10	29	48	67
1.3	-15	15	45	75	105
2.1	-14	14	42	70	98
2.2	-15	15	45	75	105
2.3	-15	14	43	72	101
3.1	-15	13	41	69	97
3.2	-13	11	35	59	83
3.3	-12	10	32	54	76
4.1	-14	14	42	70	98
4.2	-13	11	35	59	83
4.3	-11	10	31	52	73
5.1	-13	12	37	62	87
5.2	-9	9	27	45	63
5.3	-9	9	27	45	63
6.1	-14	13	40	67	94
6.2	-11	11	33	55	77
6.3	-12	11	34	57	79
7.1	-15	15	45	75	105
7.2	-9	11	31	51	71
7.3	-9	9	27	45	62
8.1	-12	12	36	60	83
8.2	-9	9	27	45	63
8.3	-13	15	43	71	99
9.1	-8	10	28	46	64
9.2	-9	9	27	45	63
9.3	-13	14	41	68	95

A Water-temp-pro temperature logger (Onset Computer Corporation, Bourne, MA) and water level logger (Global Water, Gold River, CA) were installed prior to this study at a well-mixed location approximately 10 m downstream of the study site. The temperature sensor and water level logger were installed on the bed surface in the middle of the river. The temperature sensor has a resolution of 0.02°C and an accuracy of 0.2°C , and the vented Global Water WL16 level logger has a resolution of 0.1% full scale and an accuracy of 0.1% full scale.

Numerical Methods for Groundwater Velocity Estimation

The one-dimensional, thermal convective-diffusive governing equation, assuming no internal generation or loss of heat and local thermal equilibrium between the solid-fluid matrix (Domenico and Schwartz 1990), is

$$k \frac{\partial^2 T}{\partial z^2} - nc_w \rho_w v \frac{\partial T}{\partial z} = c\rho \frac{\partial T}{\partial t} \quad (1)$$

where k is the thermal conductivity of the rock-fluid matrix ($\text{cal/s cm } ^{\circ}\text{C}$), T is the temperature ($^{\circ}\text{C}$), z is the vertical distance with a positive downward convention (cm), n is the sediment porosity (dimensionless), v is the

groundwater velocity (cm/s), c_w is the heat capacity of the fluid (cal/g °C), ρ_w is the density of the fluid (g/cm³), c is the heat capacity of the solid-fluid matrix (cal/g °C), and ρ is the wet bulk density of solid-fluid matrix (g/cm³). The assumption that the heat and flow fields can be treated as one-dimensional appears to be acceptable for conditions where vertical groundwater fluxes are relatively high and the degree of heterogeneity in physical properties is relatively low (Ferguson and Bense 2011).

Second-order, centered-difference in space and first-order implicit in time finite difference approximations were used to solve Equation 1 using the computer software MATLAB. Dirichlet boundary conditions were applied to the upper ($z = z_u$) and bottom ($z = z_b$) boundaries, as in

$$\begin{aligned} T(z_u, t) &= T_1^\ell \\ T(z_b, t) &= T_5^\ell \end{aligned} \quad (2)$$

using the upper-most and bottom-most temperature measurements at a particular observation time step (T_1^ℓ and T_5^ℓ , respectively). A Levenberg-Marquardt algorithm was used to attain best fit estimates of groundwater velocity by minimizing the sum of the squares of the residuals (SSR) between simulated vertical temperature profiles and the remaining temperature observations (T_2 , T_3 , and T_4), as in

$$\text{find } v \text{ such that } \min \sum_{i=2,3,4} (T_i^{\ell, \text{obs}} - T_i^{\ell, \text{sim}})^2 \quad (3)$$

for each observation time step, where the superscripts obs and sim refer to observed and simulated temperatures, respectively. The average error (°C) for each observation time step at each well is defined as

$$e_{t,w} = \frac{1}{n_{\text{obs}}} \sqrt{\sum_{i=2,3,4} (T_i^{\ell, \text{obs}} - T_i^{\ell, \text{sim}})^2} \quad (4)$$

where $e_{t,w}$ is the average error at a well and time step and $n_{\text{obs}} = 3$ is the number of observations at a well.

Equation 1 was solved at each time step $\Delta t = \Delta t_{\text{obs}}/m$, where Δt_{obs} is the interval between temperature measurements (3 h) and $m > 1$ is the internal time step divider, in order to improve the performance of the numerical solution as the Dirichlet conditions (Equation 2) changed. Dirichlet boundary conditions were also applied to internal time steps and were estimated by linearly interpolating the boundary temperatures between the corresponding observation time steps, Δt_{obs} . The linear interpolation of boundary conditions enhanced the performance of the numerical solution during periods of large diurnal surface-water temperature fluctuations, and hence, large fluctuations of the Dirichlet boundary conditions. Validation of the numerical solution was performed by comparison to the analytical solution from Bear (1972) for the conservative form of the advective-dispersive transport equation with a Dirichlet upstream boundary condition and a no-flux downstream boundary condition.

A spin-up approach was implemented for the purpose of simulating prior temperature conditions and used as the initial condition for model simulations. Spatial discretizations (Δz) and time steps were set to achieve values within the acceptable range of stability constraints defined by the Peclet ($Pe = v\Delta z/k$) and Courant numbers ($Cr = v\Delta t/\Delta z$), that is, $Pe \leq 2$ and $Cr \leq 1$. A spatial discretization of $\Delta z = 1$ cm and internal time step divider of $m = 2700$ were used for each simulation. The Peclet and Courant number constraints constrained the velocity estimates to $0.023 \text{ cm/s} \leq v \leq -0.023 \text{ cm/s}$. All temperature measurements that produced velocities outside this range were rejected.

Average dry bulk density (1.34 g/cm³) and porosity (0.314) of the bed sediments were determined using a graduated cylinder technique based on eight replicate samples from three locations within the study site (Tan 1995). Thermal conductivity (k) and volumetric heat capacity ($c\rho$), the effective heat capacity of the solid-fluid matrix per unit bulk volume (cal/cm³ °C) were estimated from empirical relationships found in the literature (Lapham et al. 1989), and resulting in the following parameter values: $k = 0.0036$ (cal/s cm °C), and $c\rho = 0.68$ (cal/cm³ °C). Fluid density $\rho_w = 0.999$ (g/cm³) and heat capacity $c_w = 1.00$ (g/cm³) values were determined from temperature-based formulations found in the literature (Snoeyink and Jenkins 1980). We assumed that these parameters are constant in space and time.

Results and Discussion

An average of approximately 12,000 temperature measurements, taken at 3-h intervals, was recorded in each monitoring well over the course of the study period. Large overbank flood events, instrumentation errors, and the inability to retrieve data during fall spawning periods (i.e., so as not to disturb spawning redds) resulted in the loss or fragmentation of temperature data from particular monitoring wells during specific portions of the study period. Data from monitoring wells 1.2, 3.1, 4.1, 9.1, and 9.2 were completely lost. Approximately 45,000 groundwater velocities were fitted over the study period from each of the remaining 22 monitoring wells, with 1.1% of the velocity estimates being eliminated as a result of Peclet or Courant number constraints.

Temperatures

Table 2 summarizes the temperature data collected at each well for the deepest temperature logger and in the downstream river temperature measurement location with the number of observations, averages, and standard deviations of measured temperatures over the given observation periods. Figure 5 shows a kriged map of the coefficients of variation of temperatures for the deepest temperature logger in each of the 22 monitoring wells and in the downstream river temperature measurement location. For comparison, the coefficient of variation in river temperatures during the study period was 1.03. The results in Figure 5 show that temperatures measured in the deepest

Table 2
Temperatures (in °C) Measured in Deepest Temperature Logger

Monitoring Well	Located in Spawning (S) or Nonspawning (NS) Area	Period of Observations	Number of Observations	Average Temperature	Standard Deviation of Temperature
1.1	NS	October 15, 2007 to October 1, 2008	2686	6.83	1.91
1.3	NS	June 14, 2008 to October 1, 2008	2693	6.48	3.75
2.1	NS	October 15, 2007 to October 1, 2008	2685	6.93	2.20
2.2	NS	October 15, 2007 to November 19, 2008	2932	6.62	3.37
2.3	NS	October 15, 2007 to October 1, 2008	2685	6.48	2.68
3.2	S	June 14, 2008 to October 1, 2008	878	8.34	0.72
3.3	NS	October 15, 2007 to October 1, 2008	2598	6.24	2.66
4.2	S	June 14, 2008 to October 1, 2008	878	7.83	0.76
4.3	NS	October 15, 2007 to November 19, 2008	3077	6.34	3.30
5.1	NS	October 15, 2007 to November 19, 2008	3085	6.65	3.06
5.2	S	June 14, 2008 to October 1, 2008	879	7.12	0.53
5.3	S	October 15, 2007 to November 19, 2008	2652	5.60	0.62
6.1	NS	October 15, 2007 to October 1, 2008	2624	5.98	3.66
6.2	S	June 14, 2008 to October 1, 2008	879	5.90	0.34
6.3	S	October 15, 2007 to November 19, 2008	2396	5.68	0.53
7.1	NS	October 15, 2007 to October 1, 2008	2694	6.52	3.81
7.2	S	June 14, 2008 to October 1, 2008	879	7.23	0.72
7.3	NS	October 15, 2007 to November 19, 2008	3085	6.21	0.72
8.1	NS	October 15, 2007 to October 1, 2008	2530	7.34	4.16
8.2	S	June 14, 2008 to October 1, 2008	879	8.69	1.04
8.3	NS	October 15, 2007 to November 19, 2008	3085	6.86	1.85
9.3	NS	October 15, 2007 to November 19, 2008	3083	6.41	0.78
Average		NA	2088	6.96	2.01
River		October 15, 2007 to October 1, 2008	16,943	7.05	7.24
		June 14, 2008 to October 1, 2008	7656	12.5	4.95
		October 15, 2007 to November 19, 2008	19,297	6.96	6.57

position in the wells in the spawning area are substantially less variable than wells in the nonspawning area and in the river, as measured by the coefficient of variation. For the most part, the temperature variation exhibited in the wells in the nonspawning area, as measured by the coefficient of variation, approaches the variation in temperatures exhibited in the river. These results indicate that subsurface temperatures in the spawning area are influenced significantly less by temperatures in the river, implying that upward groundwater fluxes in the spawning area are greater than those in the nonspawning area.

The coefficient of variation in monitoring wells 7.3 and 9.3 in the nonspawning area is relatively small, implying an exception to the general result of higher temperature variability in the nonspawning area. However, these sites were shallow with accumulations of fine sediments and detritus layers, which suggest poor spawning habitat conditions. Well 7.3 was located under a dense accumulation of alder branches so if spawning occurred there, it would have been difficult to detect.

One representative well from the spawning area (MW 6.3) and one representative well from the nonspawning area (MW 8.1) were selected to demonstrate temperature variation with time over the period June 14, 2008 to November 14, 2008. Figure 6a and 6b shows

temperature as a function of depth below the river bed and time at these monitoring wells, along with the river temperature measured in the upstream temperature logger. Temperature data from June 2008 to November 2008 at five vertical locations in each monitoring well were contoured using the kriging method to display results in Figure 6a and 6b.

The river temperatures in Figure 6a and 6b show pronounced diurnal and seasonal fluctuations. Figure 6a indicates that groundwater temperatures were only minimally impacted by river temperatures in the representative spawning area well, confirming the general results in Figure 5 for the spawning area wells. Figure 6b, on the other hand, indicates that for the representative nonspawning area well, fluctuations in groundwater temperatures below the riverbed closely correlate to fluctuations in river temperatures, confirming the assessment of the data in Figure 5. This correlation implies that river water was seeping downward into the river bed or that upward groundwater fluxes were low enough such that heat transfer was dominated by conduction from the streambed boundary downward into the subsurface.

Based upon the results in Figures 5, 6a, and 6b, it is likely that either riverbed hydraulic conductivities are variable or that hydraulic gradients in the upslope

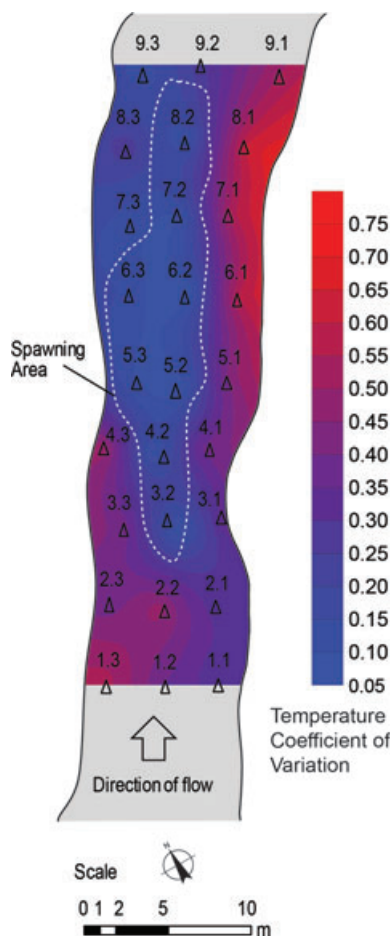


Figure 5. Kriged map of coefficients of variation of temperature from the deepest subsurface temperature loggers.

areas adjacent to the reach produce higher groundwater fluxes in the spawning section of the study site. In particular, the results in Figure 5 show that subsurface flux conditions remain relatively constant over length scales of tens of meters in the direction parallel to flow and meters in the cross-flow direction. This pattern may indicate depositional features in riverbed materials that have contrasting hydraulic conductivities. Further data collection, such as measurement of bed material hydraulic conductivities and implementation of a riparian and upland piezometer network surrounding the study site, would enable local hydrogeologic conditions to be more precisely determined.

Estimated Velocities

The error associated with model fits (Equation 4) averaged over the study period ranged between 0.05 and 0.2 °C. Comparison of the observed temperatures and the simulated vertical temperature profiles as a function of depth at various times from representative monitoring wells from the spawning area (MW 6.3) and the non-spawning area (MW 8.1) are presented in Figure 7a and 7b. Observed and simulated temperatures displayed in Figure 7a and 7b as a function of depth below the streambed represent mid-afternoon temperatures present

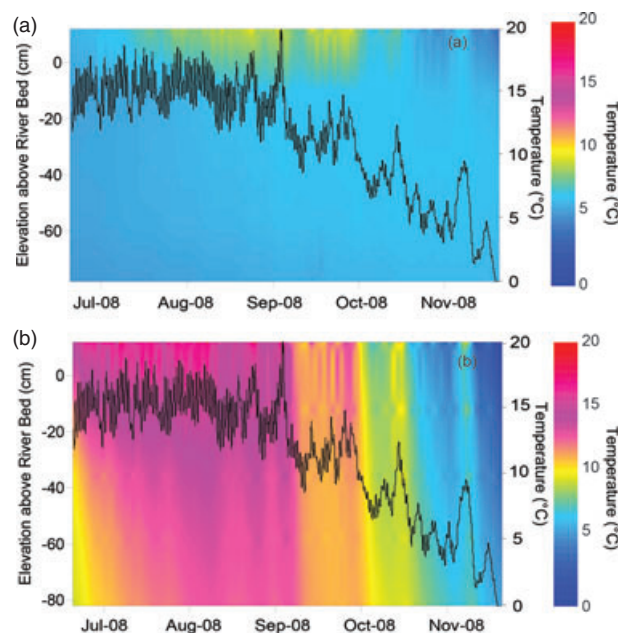


Figure 6. Temperature as a function of time and depth below river bed (color plot) and temperature in the river as a function of time (line plot) for (a) monitoring well 6.3 and (b) monitoring well 8.1.

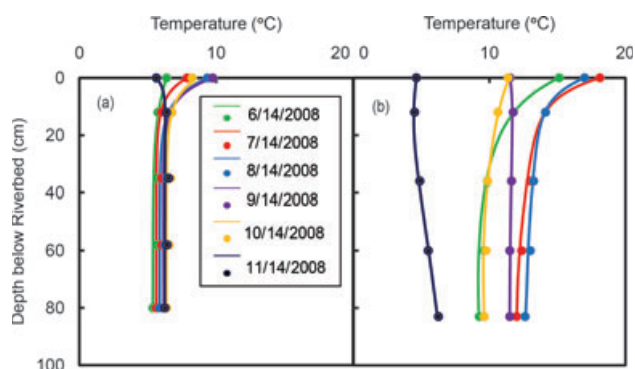


Figure 7. Observed (dots) and simulated (lines) vertical temperature profiles as a function of time at (a) monitoring well 6.3 and (b) monitoring well 8.3.

on the 14th of each month, from June through November 2008. Figure 7a, which displays temperatures for a spawning area well, shows that groundwater temperatures were not influenced by river temperatures that varied with time in this well location. The temperatures displayed in Figure 7b for a nonspawning area monitoring well, however, show marked influences from the overlying river temperatures. Figure 7a and 7b also demonstrated that model calibrations were accurate for a wide range of conditions.

Velocity estimates from all wells are summarized in Table 3 as averages and Figure 8 as frequency distributions. Note that groundwater velocities are reported as positive downward. In Figure 8, blue shades represent negative groundwater velocities, and red shades represent positive groundwater velocities. Darker colors

Table 3
Estimated Velocities

Monitoring Well	Located in Spawning (S) or Non-spawning (NS) Area	Velocity (cm/s)	
		Average	Standard Deviation
1.1	NS	-3.7×10^{-3}	1.0×10^{-2}
1.3	NS	-3.1×10^{-4}	8.6×10^{-4}
2.1	NS	4.1×10^{-4}	6.8×10^{-3}
2.2	NS	-8.2×10^{-4}	1.1×10^{-2}
2.3	NS	-5.2×10^{-4}	7.3×10^{-3}
3.2	S	-1.1×10^{-3}	1.5×10^{-3}
3.3	NS	-7.1×10^{-4}	1.1×10^{-2}
4.2	S	-6.6×10^{-3}	9.6×10^{-3}
4.3	NS	-4.6×10^{-4}	5.6×10^{-3}
5.1	NS	-7.9×10^{-4}	7.1×10^{-3}
5.2	S	-9.1×10^{-3}	1.3×10^{-2}
5.3	S	-6.8×10^{-3}	7.2×10^{-3}
6.1	NS	-7.2×10^{-4}	1.2×10^{-3}
6.2	S	-7.4×10^{-4}	2.2×10^{-3}
6.3	S	-6.4×10^{-3}	1.0×10^{-2}
7.1	NS	-2.2×10^{-4}	5.6×10^{-3}
7.2	S	-6.3×10^{-3}	1.1×10^{-2}
7.3	NS	-2.1×10^{-3}	8.7×10^{-3}
8.1	NS	-4.7×10^{-4}	3.0×10^{-3}
8.2	S	-3.6×10^{-4}	1.1×10^{-3}
8.3	NS	-5.2×10^{-3}	1.0×10^{-2}
9.3	NS	-7.2×10^{-4}	8.2×10^{-3}
Average	S	-4.7×10^{-3}	7.4×10^{-3}
	NS	-4.3×10^{-4}	7.8×10^{-3}

represent larger velocity magnitudes. The estimated velocities in Table 3 and Figure 8 indicate that most wells in the spawning area (MW 3.2, MW 4.2, MW 5.2, MW 5.3, MW 6.2, MW 6.3, MW 7.2, and MW 8.2) exhibit primarily upward (negative) velocities as indicated by the velocity frequency distribution and average velocities, with the exception of well MW 6.2 and MW 8.2. However, temperatures in the deepest measurement point in wells MW 6.2 and MW 8.2 were relatively consistent (Figure 5), which implies that groundwater velocities at these location apparently were high enough, in the upward direction, to counter temperature influences from the river.

Most wells in the nonspawning areas (MW 1.2, MW 1.3, MW 2.1, MW 2.2, MW 2.3, MW 3.1, MW 3.3, MW 4.1, MW 4.3, MW 5.1, MW 6.1, MW 7.1, MW 7.3, MW 8.1, MW 8.3, MW 9.1, MW 9.2, and MW 9.3) exhibit primarily low upward or downward velocities. The exceptional well, MW 1.1, exhibits higher upward velocities; however, it was observed that the areas surrounding MW 1.1 exhibited less than ideal spawning habitat features, such as the presence of fine sediments on the streambed, as opposed to significantly coarser material in the other well locations.

The aerial distributions of estimated groundwater velocities in Figure 8 tend to mimic the temperature distributions shown in Figure 5, where there appear to be

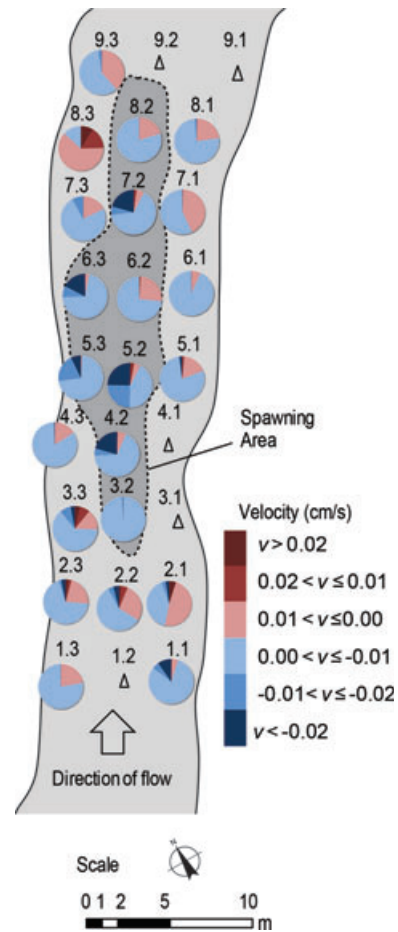


Figure 8. Frequency of estimated velocities at each monitoring well, where the size of each pie chart sector is proportional to the frequency that a velocity was found within the given range. Blue colors represent upward (negative) velocities and red colors represent downward (positive) velocities. Each pie chart is centered at the location of the corresponding monitoring well.

continuous “patches” of consistent upward or consistent downward velocities. As mentioned previously, this pattern may be the result of depositional features in the riverbed materials that produce continuous distributions of hydraulic conductivities in those materials.

Estimates of groundwater velocities and river stage as a function of time from representative monitoring wells from the spawning area (MW 6.3) and the nonspawning area (MW 8.1) are presented in Figure 9a and 9b, respectively. All figures contain groundwater velocity estimates for the entire study period except for the spring of 2008. Freeze-thaw cycles in the river removed MW 6.3 from the subsurface in February 2008 and MW 8.1 in March 2008. All temperature loggers were removed from the study site in May 2008 and re-installed in June 2008. Figure 9a and 9b shows that the velocities estimated from the spawning area (MW 6.3) are more consistently upward and greater in magnitude than the velocities estimated from temperature data in the nonspawning area (MW 8.1).

Comparison of the velocity and stage data for monitoring well 6.3 indicates potential seasonal correlations

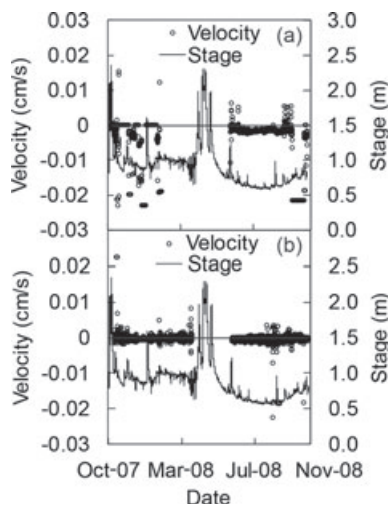


Figure 9. Estimated velocities (symbols) at (a) monitoring well 6.3 and (b) monitoring well 8.1 and river stage (lines) as a function of time.

associated with the unseasonably large fall 2007 flood events, the post snowmelt spring 2008 falling limb, summer 2008 baseflow period, and fall 2008 rising limb. The unseasonably large fall 2007 flood events, the post snowmelt spring falling limb, and fall 2008 rising limb each coincide with periods of increased groundwater velocity fluctuations. During the fall spawning period of 2007 and 2008, groundwater velocities were greater in magnitude and predominantly negative. During periods of stable hydrologic conditions, such as the summer 2008 baseflow period, groundwater velocities also tended to be stable and were predominantly negative (upward). Groundwater velocity fluctuations in monitoring well 8.1 do not appear to coincide with seasonal hydrologic conditions.

Conclusions

A monitoring well system was installed to measure subsurface temperatures underneath a riverbed over a 13-month period. In total, over 200,000 temperature measurements were recorded at five vertical locations in 22 monitoring wells distributed regularly over approximately 80-m of river channel. The temperature data were inverted to obtain subsurface groundwater velocities using a numerical approximation of the heat transfer equation applicable to fluids in porous media. Approximately 45,000 values of groundwater velocities were estimated.

Although the study area was relatively homogenous with respect to surficial substrate and habitat characteristics, coaster brook trout have been consistently observed to return to spawn in specific areas within the 80-m river length study area. The monitoring well locations were defined as being either inside or outside the areas where spawning behavior occurred. Temperatures within the stream substrate in the spawning area were generally less variable than river temperatures, indicating that

groundwater velocities in this area were high enough to minimize within-substrate mixing of river and groundwater. Substrate temperatures in the nonspawning area were generally more variable, and closely tracked river temperatures as they varied temporally.

Estimated velocities in the spawning and nonspawning areas confirmed that groundwater velocities in the spawning area were primarily in the upward direction. In the nonspawning area, groundwater velocities were mostly either in the downward direction or, if they were in the upward direction, the magnitude of the average velocity was generally lower. To the best of our knowledge, no other study has implemented a high temporal and spatial resolution temperature network to quantify subsurface water velocities over long time periods to evaluate a similar ecological hypothesis. The overall result of this work is to point out the significance of groundwater seepage in the distribution of spawning activities by coaster brook trout, which is critical for identifying river reaches with high priority for protection or restoration for coaster brook trout rehabilitation.

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